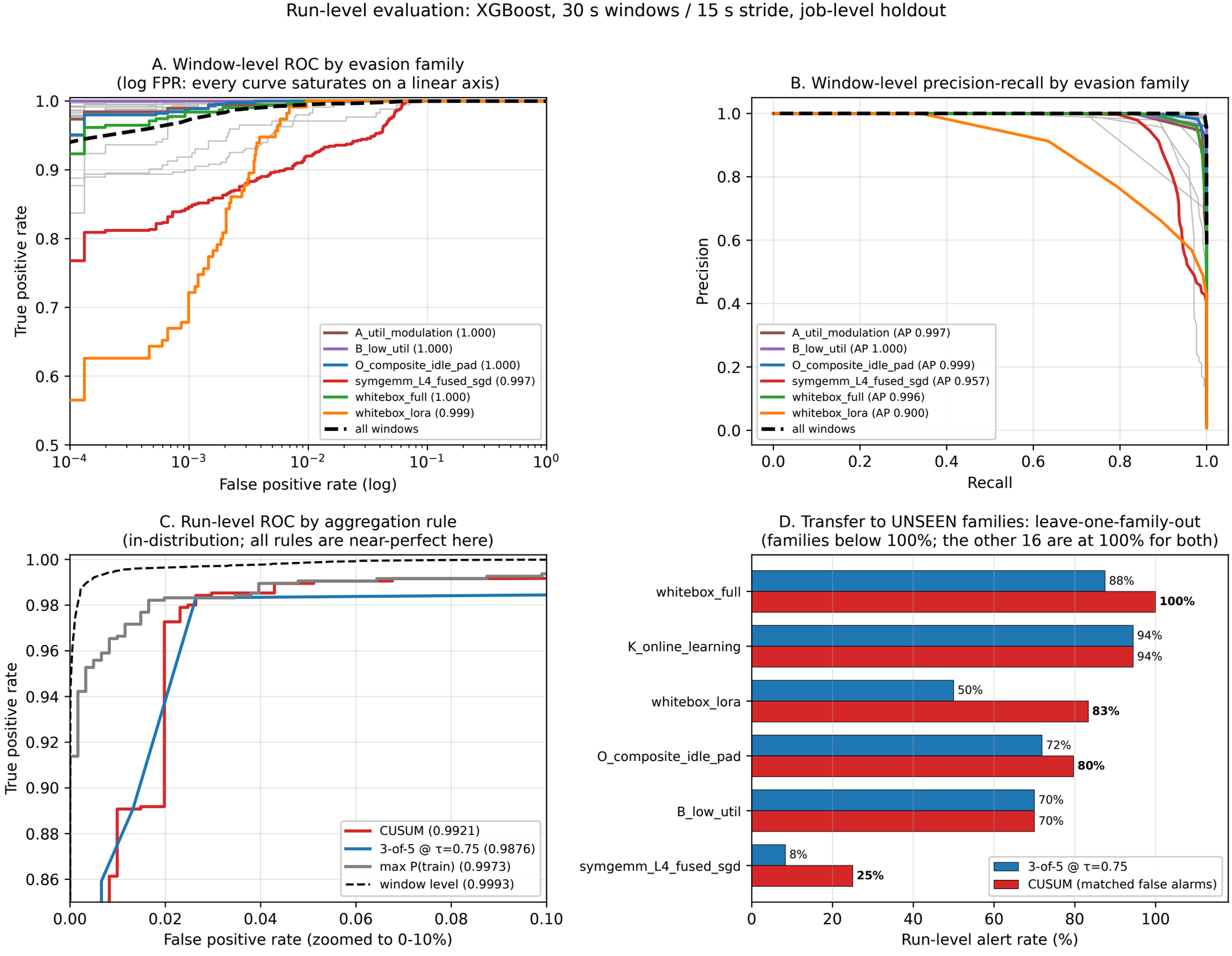
Run-level evaluation: XGBoost, 30 s windows / 15 s stride, job-level holdout
A. Window-level ROC by evasion family
(log FPR: every curve saturates on a linear axis)
True positive rate
False positive rate (log)
A_util_modulation (1.000)
B_low_util (1.000)
O_composite_idle_pad (1.000)
symgemm_L4_fused_sgd (0.997)
whitebox_full (1.000)
whitebox_lora (0.999)
all windows
B. Window-level precision-recall by evasion family
Precision
Recall
A_util_modulation (AP 0.997)
B_low_util (AP 1.000)
O_composite_idle_pad (AP 0.999)
symgemm_L4_fused_sgd (AP 0.957)
whitebox_full (AP 0.996)
whitebox_lora (AP 0.900)
all windows
C. Run-level ROC by aggregation rule
(in-distribution; all rules are near-perfect here)
True positive rate
False positive rate (zoomed to 0-10%)
CUSUM (0.9921)
3-of-5 @ τ=0.75 (0.9876)
max P(train) (0.9973)
window level (0.9993)
D. Transfer to UNSEEN families: leave-one-family-out
(families below 100%; the other 16 are at 100% for both)
whitebox_full
K_online_learning
whitebox_lora
O_composite_idle_pad
B_low_util
symgemm_L4_fused_sgd
88%
100%
94%
94%
50%
83%
72%
80%
70%
70%
8%
25%
3-of-5 @ τ=0.75
CUSUM (matched false alarms)
Run-level alert rate (%)